

Course Code: CST306

Course Name: ALGORITHM ANALYSIS AND DESIGN

Max. Marks: 100

Duration: 3 Hours

**PART A***Answer all questions, each carries 3 marks.*

Marks

- |    |                                                                                                                                                               |     |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1  | What are the characteristics of a good algorithm?                                                                                                             | (3) |
| 2  | Solve $T(n)=4T(n/2) + n^3$ using master method.                                                                                                               | (3) |
| 3  | Can we use DFS to detect cycles in a graph? Justify your answer.                                                                                              | (3) |
| 4  | Define AVL tree. What is the advantage of AVL tree? Give an example                                                                                           | (3) |
| 5  | Strassen's multiplication method is used to multiply two $n \times n$ matrices when $n$ is a power of 2. How it can be modified when $n$ is not a power of 2? | (3) |
| 6  | Define spanning tree of a graph. Write the total number of spanning trees possible for a complete graph with 4 vertices.                                      | (3) |
| 7  | Write a recurrence to represent the number of ways to parenthesize a chain of $n$ matrices.                                                                   | (3) |
| 8  | Define Travelling Salesman problem.                                                                                                                           | (3) |
| 9  | What do you mean by tractable problem? Give an example.                                                                                                       | (3) |
| 10 | Define graph colouring problem.                                                                                                                               | (3) |

**PART B***Answer one full question from each module, each carries 14 marks.***Module I**

- |    |                                                                                 |     |
|----|---------------------------------------------------------------------------------|-----|
| 11 | a) Solve the following recurrence using iteration method.<br>$T(n)=2T(n/2) + n$ | (6) |
|    | b) Define the asymptotic notations: Big Oh, Big Omega, Theta and little omega.  | (8) |

**OR**

- |    |                                                                                                |     |
|----|------------------------------------------------------------------------------------------------|-----|
| 12 | a) Illustrate best case, average case and worst-case complexity with insertion sort algorithm. | (7) |
|    | b) Solve the following recurrence using recursion tree method.<br>$T(n)=T(n/3) + T(2n/3)+cn$   | (7) |

**Module II**

- 13 a) Explain the different operations possible on disjoint sets. Implement UNION using linked list representation of disjoint sets. (7)
- b) Give Breadth First Search algorithm for graph traversal. Perform its complexity analysis. (7)

**OR**

- 14 a) Write an algorithm to find strongly connected components of a graph. Illustrate with an example. (7)
- b) Give Depth First Search algorithm for graph traversal. How the edges of a graph are classified based on DFS? (7)

**Module III**

- 15 a) Write the control abstraction for Greedy design technique. Give a greedy algorithm for fractional knapsack problem. (7)
- b) Illustrate the divide and conquer approach by applying 2 way merge sort for the input array: [15,12,14,17,11,13,12,16]. Write the recurrence for merge sort and give the complexity. (7)

**OR**

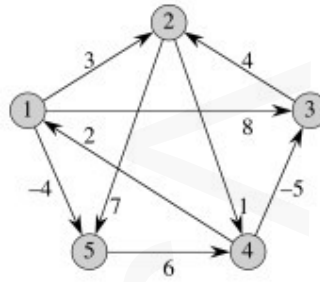
- 16 a) Find an optimal solution to the fractional knapsack instance  $n=5$ ,  $m=60$ ,  $(p_1, p_2, \dots, p_5) = (30, 20, 100, 90, 160)$  and  $(w_1, w_2, \dots, w_5) = (5, 10, 20, 30, 40)$ , Where  $m$  is the knapsack capacity,  $p_i$  is the profit of  $i$ th item and  $w_i$  is the weight of  $i$ th item. (8)
- b) Write Dijkstra's algorithm for single source shortest path. Perform its complexity analysis. (6)

**Module IV**

- 17 a) Find an optimal paranthesization of a matrix-chain product whose sequence of dimensions is  $4 \times 10, 10 \times 3, 3 \times 12, 12 \times 20, 20 \times 7$  using dynamic programming (8)
- b) Explain how Travelling Salesman Problem can be solved using Branch and Bound method. (6)

**OR**

- 18 a) Explain backtracking technique. How 4-queens problem can be solved using backtracking? (6)
- b) Explain Floyd-Warshall Algorithm. Using the algorithm find all pair of shortest paths in the following graph. (8)



### Module V

- 19 a) Discuss the advantages of randomized algorithms over deterministic algorithms. (7)  
 Discuss Las Vegas and Monte Carlo algorithms with a suitable example.
- b) Give a randomized version of quicksort algorithm and perform its expected running time analysis. (7)

OR

- 20 a) Define bin packing problem. Discuss the first fit strategy for solving it. State the approximation ratio of the algorithm. (7)
- b) Prove that vertex cover problem is NP Complete. (7)

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